A Multi-Source Retrieval-Augmented Generation Framework for Reliable Medical Question Answering

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***Abstract*—This document is a model and instructions for LaTeX.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

This document is a model and instructions for LaTeX. Please observe the conference page limits.

II. Related Work

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III. Methodology

We introduce the methodology here.

A. Problem Formulation

Medical question-answering is a critical real-world problem because incorrect or unsupported answers may lead to harmful clinical decisions. Prior approaches to this problem fall into two main categories: (i) LLM-only generation, which relies primarily on intrinsic knowledge embedded within large language models (LLMs) and is prone to hallucinations or inaccuracies due to limited medical domain expertise, and (ii) single-source Retrieval-Augmented Generation (RAG) systems, which retrieve and condition on a single external corpus. While these approaches improve accuracy, they face limitations, such as (a) a lack of context-sensitive source planning when querying multiple heterogeneous knowledge bases, (b) a failure to provide multi-level verification (automated cross-checks, expert validation), and (c) limited transparency regarding the provenance of generated answers. Our system aims to address these shortcomings by (1) retrieving from multiple complementary sources, (2) using a context-aware integrator to align and fuse heterogeneous evidence, and (3) applying hierarchical verification, culminating in a human expert oracle when necessary [1].

Let $\mathcal{Q}$ denote the space of natural-language medical queries and $\mathcal{A}$ the space of candidate answers. Let the ground-truth answer for a query $q \in \mathcal{Q}$ be $a^\star(q) \in \mathcal{A}$. We consider three complementary knowledge sources $\mathcal{K} = \{K_{\text{int}}, K_{\text{dom}}, K_{\text{ext}}\}$, where K_{int} denotes the LLM's intrinsic (pretrained) knowledge, K_{dom} a curated, expert-annotated medical QA dataset, and K_{ext} external online sources (e.g., open web pages, guideline repositories, and open-source LLM outputs). For each structured source K_i we assume a document space $\mathcal{D}_i$ and define a retriever, such as:

$$\mathcal{R}_i : \mathcal{Q} \to 2^{\mathcal{D}_i} \tag{1}$$

which returns a finite set of top-k passages $\mathcal{R}_i(q) \subseteq \mathcal{D}_i$. The integrator is a function

$$\mathcal{I} : \prod_i 2^{\mathcal{D}_i} \times \mathcal{H} \to \mathcal{D}, \tag{2}$$

which maps the retrieved passages together with the LLM intrinsic hypothesis space $\mathcal{H}$ into a fused evidence set $\mathcal{D}$ (where $\mathcal{D}$ is the set of documents used by the reader). The reader/generator is

$$G : \mathcal{Q} \times \mathcal{D} \times \mathcal{H} \to \mathcal{A}, \tag{3}$$

which produces an answer $\hat{a} = G(q, \mathcal{D}, h_0)$ given the fused evidence $\mathcal{D}$ and an intrinsic hypothesis $h_0 \in \mathcal{H}$ (the LLM's prior output for q). Finally, a verifier (automatic checks and human expert) is modeled as

$$V : \mathcal{A} \times \mathcal{D} \to \{\text{accept}, \text{reject}\}, \tag{4}$$

and the system outputs $\tilde{a} = \hat{a}$ only when $V(\hat{a}, \mathcal{D}) = \text{accept}$; otherwise the system triggers a fallback (human revision or abstention). We now formalize the pipeline and give a conditional correctness guarantee. Define the union document space $\mathcal{D}^{\cup} = \bigcup_i \mathcal{D}_i$. For a fixed query q let $E^{\star}(q) \subseteq \mathcal{D}^{\cup}$ denote the set of *supporting evidence documents* whose content, when correctly interpreted, suffices to justify the ground-truth answer $a^{\star}(q)$. The pipeline executes as:

$$h_0 = \text{LLM}_{\text{pre}}(q), \tag{5}$$

$$\mathcal{D}_i = \mathcal{R}_i(q) \quad (i \in \{\text{dom}, \text{ext}\}), \tag{6}$$

and the next steps are:

$$\mathcal{D} = \mathcal{I}(\{\mathcal{D}_i\}_i, h_0), \tag{7}$$

$$\hat{a} = G(q, \mathcal{D}, h_0), \tag{8}$$

$$\tilde{a} = \begin{cases} \hat{a} & \text{if } V(\hat{a}, \mathcal{D}) = \text{accept}, \\ \text{defer / escalate} & \text{otherwise.} \end{cases} \tag{9}$$

Correctness Analysis and Reliability Guarantee: The effectiveness of the proposed multi-source RAG framework can be theoretically justified by analyzing the end-to-end conditions under which the system produces a correct and reliable answer. Let $E^{\star}(q) \subseteq \mathcal{D}^{\cup}$ denote the set of evidence documents whose content logically entails the ground-truth answer $a^{\star}(q)$ for a query q. The system's correctness depends on three fundamental properties: (i) at least one such evidence document exists in the global knowledge space, (ii) the retrieval and integration pipeline successfully includes this evidence in the fused knowledge set, and (iii) the generator and verification modules faithfully interpret and validate this evidence. When these conditions hold, we can formally establish the following guarantee.

Theorem 1 (Deterministic Correctness Guarantee)**.** *If (1) there exists at least one evidence document $d^{\star} \in E^{\star}(q)$ whose content entails $a^{\star}(q)$, (2) the retrieval and integration components ensure $d^{\star} \in \mathcal{D}$, (3) the generator G produces an answer $\hat{a}$ that is semantically consistent with $d^{\star}$, and (4) the verification function V accepts only those answers entailed by $\mathcal{D}$, then the final system output satisfies:*

$$\tilde{a} = a^{\star}(q).$$

Proof. Under condition (1), a valid evidence document $d^{\star}$ exists. Condition (2) guarantees that $d^{\star}$ is retrieved and included in the fused evidence set $\mathcal{D}$. Condition (3) ensures that the generator produces $\hat{a}$ consistent with $d^{\star}$, implying $\hat{a} = a^{\star}(q)$. Finally, condition (4) implies $V(\hat{a}, \mathcal{D}) = \text{accept}$, and thus the final output $\tilde{a}$ is identical to $a^{\star}(q)$. □

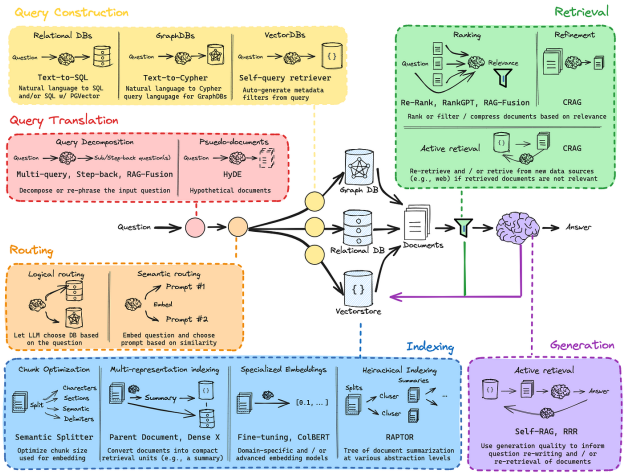

Fig. 1. A short caption of the model

The above result provides a deterministic guarantee: if each component of the pipeline behaves as specified, the system's output will always align with the ground-truth answer. In practice, however, retrieval noise, model limitations, and verification imperfections introduce stochasticity into the process. We therefore extend the analysis with a probabilistic formulation to capture real-world reliability.

Probabilistic Reliability Analysis: Let p_i denote the probability that the retriever associated with knowledge source K_i retrieves at least one evidence document $d^{\star}$ for query q. Assuming independence between retrieval events across sources, the probability that at least one evidence document is successfully retrieved is:

$$P_{\text{retrieval}} = 1 - \prod_i (1 - p_i).$$

Let p_G denote the probability that the generator produces an answer faithful to the retrieved evidence, and p_V the probability that the verifier accepts a correct evidence-supported answer. Then, the overall probability that the system produces the correct final answer is bounded below by:

$$P(\tilde{a} = a^{\star}(q)) \geq P_{\text{retrieval}} \cdot p_G \cdot p_V = \big(1 - \prod_i (1 - p_i)\big) p_G p_V.$$

This result quantifies how multi-source retrieval directly enhances system robustness: as the number of independent sources increases, the term $1 - \prod_i (1 - p_i)$ approaches 1, thereby increasing the probability that sufficient evidence is available for faithful synthesis and verification. In other words, the proposed multi-source, multi-level verification framework is not only theoretically sound under ideal conditions but also exhibits provable reliability improvements under practical, probabilistic constraints.

B. System Workflow

The proposed system integrates a multi-source retrieval-augmented generation (RAG) pipeline to deliver reliable medical question answering (QA) with multi-level verification.

As depicted in **Figure X**, the workflow consists of five key modules: Query Understanding, Knowledge Retrieval, Knowledge Integration, Multi-Level Verification, and Answer Generation. Below, we detail the function and interactions of these modules, explaining how they collaboratively work to ensure the accuracy, reliability, and transparency of the system's output.

1. Query Understanding Module

The process begins with the **Query Understanding Module**, which takes as input a medical query $q \in \mathcal{Q}$, represented as a natural language question. The primary role of this module is to parse and interpret the query using the pre-trained LLM (Large Language Model). The LLM dissects the query to identify its medical context, the type of information being requested (e.g., diagnosis, treatment plan, disease mechanism, drug interactions), and any specific constraints or details implied by the query. For instance, in a clinical setting, the query might ask for the diagnosis of a particular condition based on symptoms or request advice on the management of a specific medical procedure.

The LLM generates a context-specific hypothesis from its internal knowledge, which helps shape the subsequent retrieval process. This hypothesis may serve as an initial response or a starting point for further information extraction. The key challenge here is ensuring that the LLM accurately recognizes the intent of the query while preserving the necessary medical context, which is critical for the correctness of the final output.

2. Knowledge Retrieval Layer

Once the query is understood, the system proceeds to the **Knowledge Retrieval Layer**, which consists of three distinct retrieval processes. The first of these processes involves leveraging the **Intrinsic Knowledge** embedded in the LLM. Based on the initial hypothesis generated in the previous module, the LLM produces an answer that reflects the knowledge it has acquired during its pretraining phase. This answer, however, is not definitive and is subject to further refinement through the retrieval of more specialized information.

Next, the **Medical Knowledge Base Retrieval** step activates the system's domain-specific retrieval mechanism. Here, the system queries a curated medical question-answer dataset, constructed by medical professionals. This dataset is composed of expert-annotated medical content, including clinical guidelines, research articles, and structured medical knowledge sources. The retrieved passages from this dataset serve to anchor the system's response in evidence-based medical literature.

The third retrieval process is the **External Web Retrieval**. This component is designed to access up-to-date medical information from trusted online sources, such as medical journals, guidelines from authoritative health organizations, and open-access generative models. This step is crucial for incorporating the most recent research, treatment recommendations, or expert consensus that may not have been captured during the LLM's pretraining. It allows the system to dynamically adjust to new medical findings and provide answers that reflect current practices.

3. Knowledge Integration Layer

After gathering knowledge from the aforementioned retrieval processes, the system moves to the **Knowledge Integration Layer**. The primary function of this layer is to merge the diverse sources of knowledge into a coherent and reliable answer. The integration process is guided by a context-aware integration model that performs several key operations. First, it ensures **semantic alignment** between the retrieved knowledge sources. This step checks for consistency in terminology, medical concepts, and context across the documents retrieved from the intrinsic knowledge, domain-specific dataset, and external sources.

Next, **redundancy elimination** is performed to avoid conflicts or repetition of information. If multiple sources provide similar or identical content, the system eliminates any unnecessary duplications, leaving the most relevant and authoritative information. The integration model also ranks the evidence based on **evidence strength**—prioritizing sources that are more authoritative, recent, or directly relevant to the query. This ensures that the integrated set of documents used for answer generation is not only comprehensive but also consistent and reliable.

4. Multi-Level Verification Layer

The **Multi-Level Verification Layer** is where the system undergoes rigorous validation to ensure the correctness and safety of the generated response. This module consists of three stages: automated verification, external verification, and human-in-the-loop verification.

- **Automated Verification** involves cross-checking the retrieved evidence for internal consistency. This includes detecting contradictions within the retrieved documents, ensuring that the facts align across different sources, and verifying that the generated answer is logically consistent with the evidence. The automated checks also assess the completeness of the retrieved information, ensuring that critical pieces of evidence are not missing.
- **External Verification** is performed by referencing authoritative sources such as clinical guidelines, peer-reviewed studies, and publicly available medical databases. This ensures that the retrieved knowledge aligns with established medical standards and the latest evidence-based practices.
- **Human-in-the-Loop Verification** is the final safeguard, in which a human medical expert is consulted if the automated and external verification processes raise any doubts or uncertainties. This step is particularly important for complex or high-risk medical queries where the potential for harm due to incorrect information is significant. The expert reviews the generated answer, assesses its medical validity, and either approves or revises the response. This multi-level verification process ensures that the system's output is both accurate and safe for clinical application.

5. Answer Generation and Explanation Module

Finally, the **Answer Generation and Explanation Module** produces the system's final output. Based on the integrated and verified knowledge, the LLM generates a medically accurate

response to the user's query. This response is accompanied by **provenance metadata** that provides transparency regarding the source of the information, including citations to the retrieved documents and evidence used to support the answer. This transparency not only improves trust in the system's output but also allows healthcare professionals to audit the process and verify the reliability of the response.

The explanation module also helps to clarify the rationale behind the system's conclusions. By providing explanations alongside the generated answers, the system fosters a better understanding of the medical reasoning behind each recommendation or diagnosis. This is especially important in clinical decision-making, where healthcare providers need to make informed choices based on transparent and well-supported evidence.

REFERENCES

[1] W. Ali, R. Kumar, X. Zhou, and J. Shao, "Responsible recommendation services with blockchain empowered asynchronous federated learning," *ACM Trans. Intell. Syst. Technol.*, vol. 15, no. 4, pp. 78:1–78:24, 2024.